

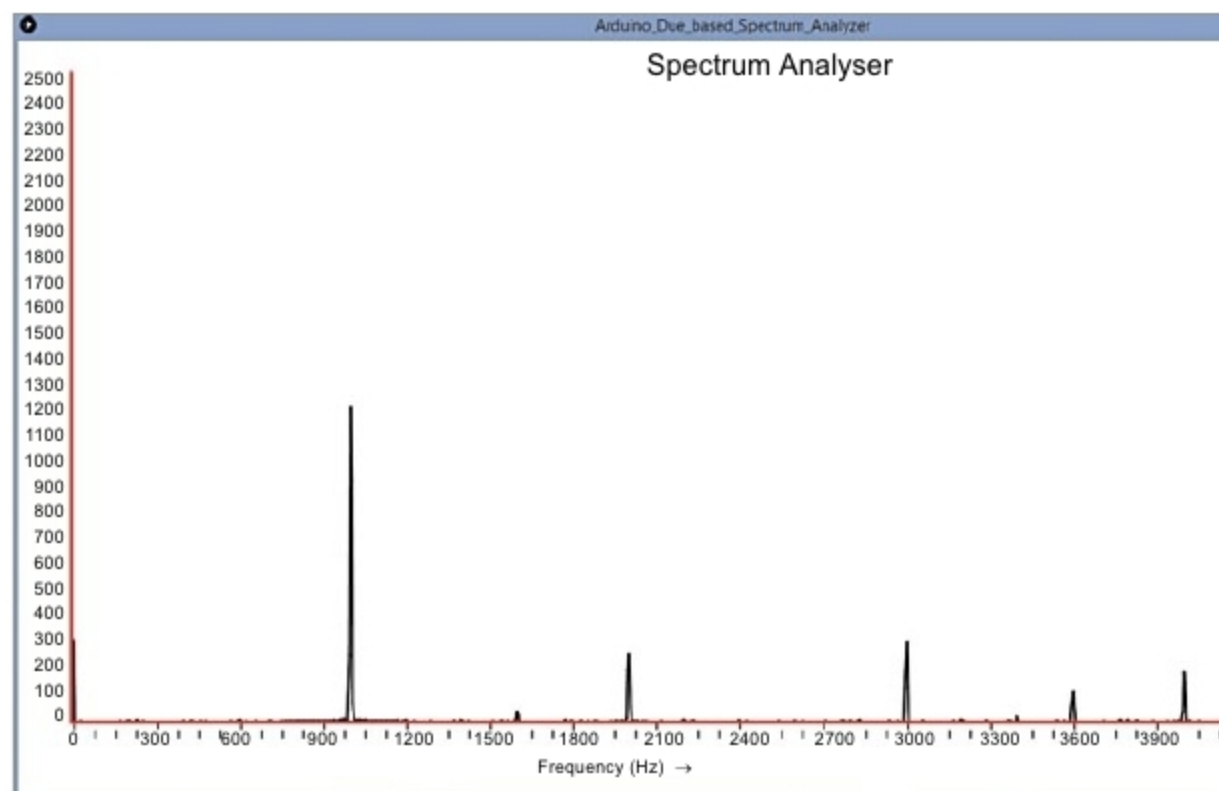


Fig. 3: Snapshot of the spectrum analyser screen for around 1kHz input signal

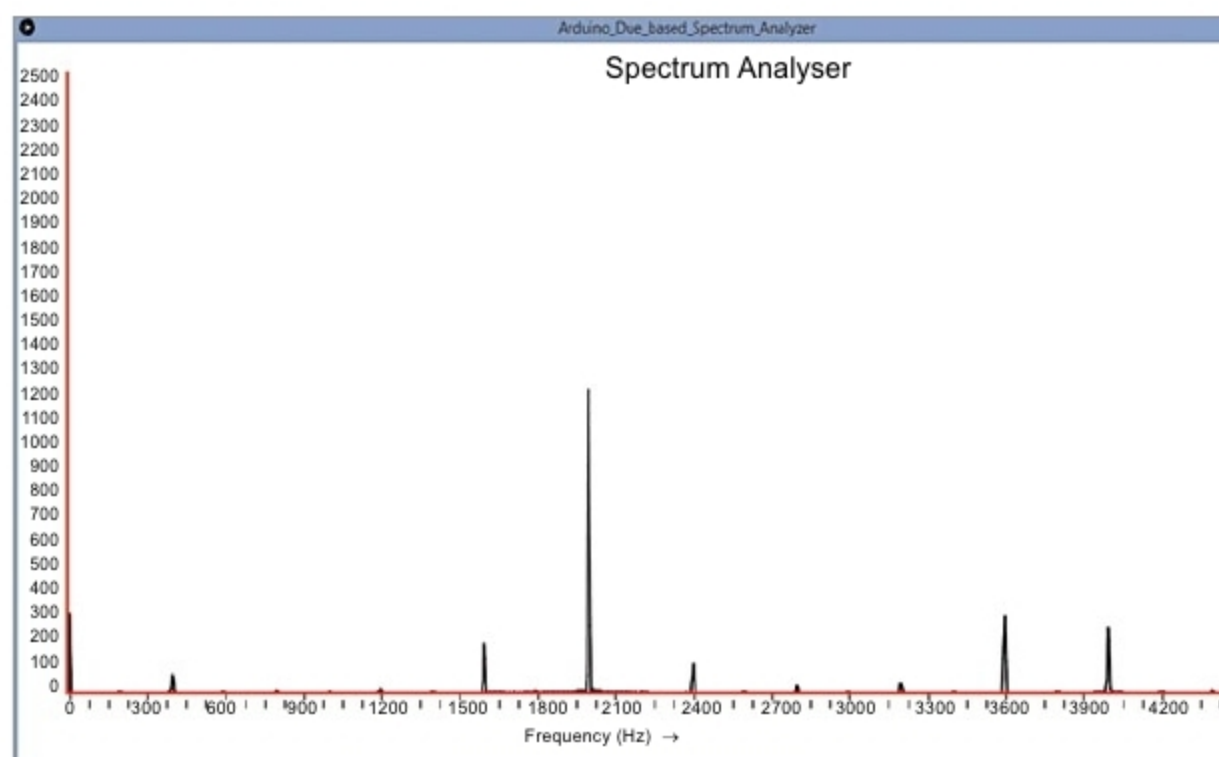


Fig. 4: Snapshot of the spectrum analyser screen for 2kHz input signal

FFT library is used for Arduino Due board.

Processing code to view the frequency spectrum plot on the laptop has been designed using Processing software. Source program (FFT_acc_Vs_k.pde) written in Processing can

be modified as per requirement.

Snapshots of the output screen for input signals with 1kHz and 2kHz are shown in Figs 3 and 4, respectively, using a 2048-point FFT with $f_s = 9615$ Hz.

This project can be easily extended

for multi-channel spectrum analysis, as Arduino Due has twelve analogue input channels.

Testing steps

The steps for testing are given below.

1. Download and install Arduino IDE from <https://www.arduino.cc/en/Main/Software>, and Processing software from <https://processing.org/download/>

2. Connect Arduino DUE board with the laptop using USB cable and open Arduino IDE. Add SplitRadixRealP library under libraries folder of the IDE. Open Arduino code (FFT_Split_for_GUI.ino) using Arduino IDE software. Under Tools, select Arduino Due (Programming port) board and COM port. Next, compile and upload the source code to Arduino DUE board.

3. Open Processing source code (FFT_acc_Vs_k.pde) using Processing software. Change COM port in the source code.

4. Connect input from a function generator to Arduino Due pin A1. Make sure that ground is common and run the program/sketch. Output spectrum window will open and show the spectrum for the given input signal.

5. You can view the spectrum in the given range on x-axis, while y-axis shows amplitude of the input signal given at A1 pin of Arduino Due board. **EFY**



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